

Food Structure, Rheology, and Texture

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Introduction – Food Structure

Food structure is the organization of food constituents at multiple spatial scales and their interactions. In simple terms, different types of food structures are formed when food ingredients are mixed together to make a food product. Many food properties relevant to process engineering and product quality are related to structure. Insights into food structure and how it changes during processing operations are essential for producing high-quality food.

Structures that are relevant to food properties can be considered at different length scales (**Figure 1**):

1. At the smallest scale is the molecular structure: chemical structures that make up the constituents such as water, minerals, vitamins, flavor components, fatty acids, lipids and protein monomers, etc.
2. The next level up is the microscopic level: aggregation of molecules and their assembly into components, colloids and networks which are typically measured between nanometers and micrometers. A range of classic supermolecular structures in food are casein micelles, milk fat globules, starch granules, biopolymer network/gels such as the gluten structure in dough, pectin gel, etc.

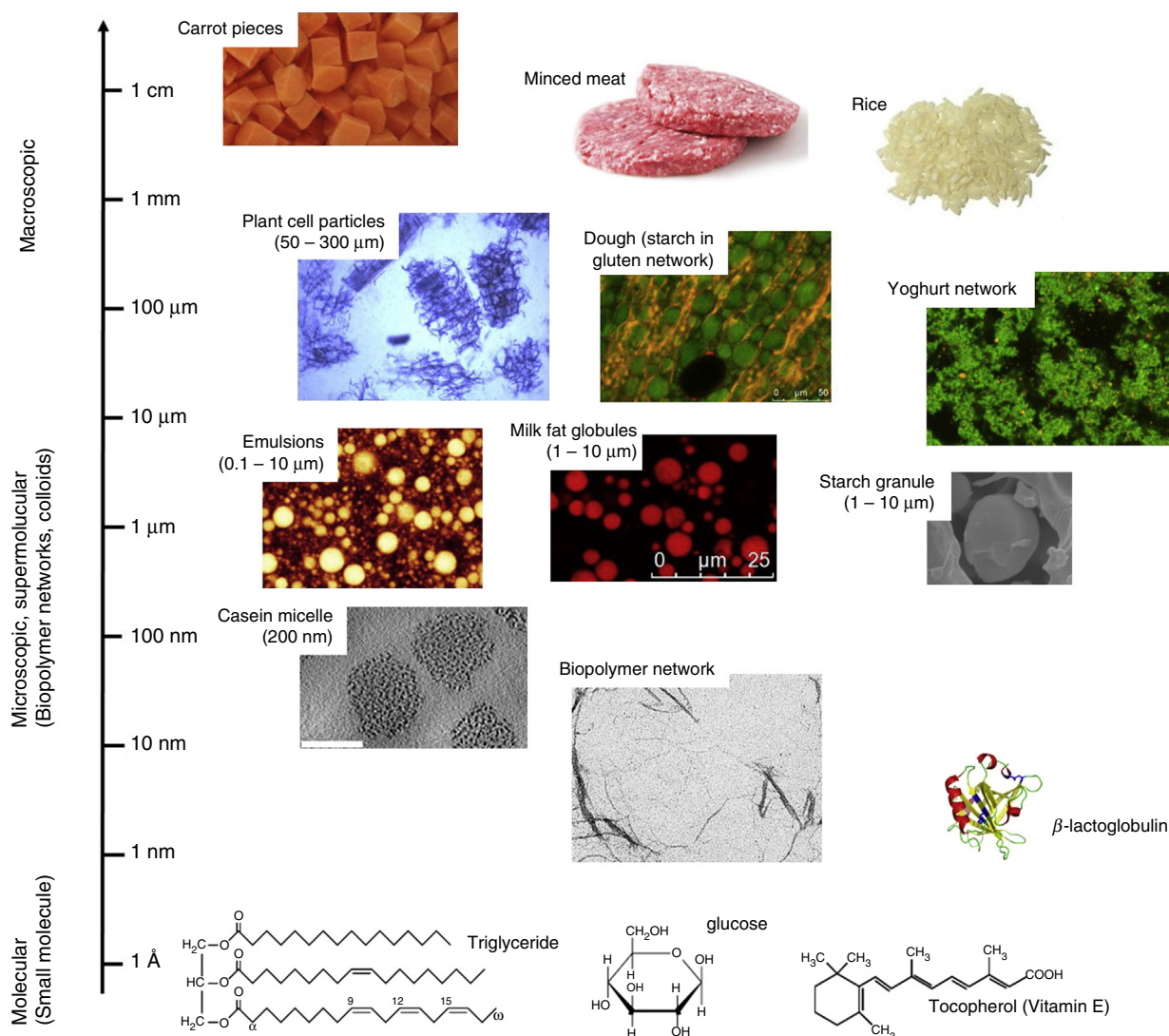


Figure 1 The length scale of structures and examples of representative molecules, food components, networks, and food structures. Microscopic images are kindly provided by Drs S.K. Øiseth, T.H. McCann and A. Leis, CSIRO (Werribee and Geelong, Australia).

3. The macroscopic level of structure considers those features that are perceived by human senses, such as texture and taste. They are often associated with specific types of foods by consumers, for example, tomato paste or fruit purees, recombined meat such as burgers and sausages, porous structures such as cakes and biscuits, and different types of grains, etc.

The processability, texture, flavor, and shelf-life qualities of food are controlled not just by chemical composition, but also by how the various ingredients are distributed and interact at the nano and microscopic length scales. Food structures vary enormously from relatively homogenous liquids to complex, multiphase colloidal systems containing fats, proteins, polysaccharides, salts, and water in the form of fibers, droplets, crystals, networks or even air bubbles. The size, shape and distribution of these structures greatly influence product stability as well as sensory properties and bioavailability of nutrients for absorption when food is consumed and digested (Turgeon and Rioux, 2011).

The important role of food structures in the area of nutrition is more and more recognized: how digestion of food macronutrients such as starch, fat and protein occurs depending on the matrices of food structure and thus accessibility of digestive enzymes in the gut. The same is true of micronutrient accessibility – both the chemical and physical states affect whether the body is able to derive the potential benefit of minerals, phytochemicals and vitamins, etc. (Parada and Aguilera, 2007).

Food Colloids

Colloids are a widely encountered structural component present in many different food systems. A colloid can be classified as a dispersion of one material state in another, in which the dispersed component typically has dimensions across the micron length scale. The material states that comprise colloids are usually called the dispersed phase and the continuous phase, and each of these can be comprised of solid, liquid, or gas. Usually, the two phases are immiscible with each other, and thus colloidal systems are thermodynamically unstable. Instead, colloids can be kinetically stabilized, either through manipulation of the dispersed phase (such as reducing particle size) or through the continuous phase (increasing viscosity, or imparting a yield stress). In this way, the physical stability (or structural lifetime) of a colloid can be controlled and that for foods can range from a few seconds to in excess of a year, depending on the product in question (Dickinson, 2015).

Emulsions

Oil and water, when combined, are well known for their immiscibility, yet both materials are frequently encountered together in many food systems. The particular colloidal state used to achieve metastability of mixture of oil and water is the emulsion, in which one phase is dispersed in the other in the form of small droplets. For food products, particularly soft solid foods, the oil-in-water state is more commonly encountered, providing a structural, material, and sensory contribution to products such as homogenized milk, yogurt, mayonnaise, ice cream, and cheese (with butter and margarine as examples of water-in-oil type emulsions). Emulsion stabilization is usually achieved through a combination of processing, in the form of homogenization, and surface stabilization of droplets through formulation design. High-pressure homogenizers and colloid mills are most frequently used in food manufacturing to produce emulsions. Both processes exert high shear forces on mixtures of oil and water, and the resulting flow properties under these shear conditions are sufficient to disperse the oil phase into small oil droplets (typically ranging from ~ 0.2 to $20\ \mu\text{m}$ in foods). Ingredient stabilization, in which surface active components in the formulation (such as proteins – notably dairy proteins, and polar lipids such as monoglycerides and lecithins) are able to adsorb at the newly formed interface, can be used to not only provide stabilization against coalescence and phase separation, but also to promote specific interactions between droplets. Emulsion structuring through controlled flocculation or partial coalescence contributes to the stability and material properties and sensory properties of many foods, such as cream cheese, which comprise a network structure consisting of protein and fat droplets, and whipped cream for which the foam structure is stabilized by a scaffold of agglomerated fat droplets.

Foams

Foams are another example of a colloidal state commonly found in foods, such as cakes, breads, whipped cream, confectionary and beverages. Foams are dispersion of air in a liquid continuous phase (which is usually water). Formation and stabilization of foams is similar to that of emulsions, requiring the generation of air bubbles as the dispersed phase (through incorporation of air under high shear, but also potentially through sparging or pressure drop) and their stabilization through adsorption of an appropriate surface active material at the air–water interface (again, proteins and polar lipids can be effective in this respect). Loss of stability occurs through phase separation (drainage), coalescence and disproportionation (in which air diffuses from small bubbles into large bubbles).

Bubble size is typically larger than that observed for emulsions (ranging from $\sim 50\ \mu\text{m}$ to several millimeters depending on product), and due to the large density difference between air and water, long-term stabilization can be challenging. For some products, such as beer foam, bubble lifetime only needs to be maintained over a matter of few minutes (although even during this short time, the foam provides both a sensory contribution and visual cue to the product). However, for other products such as ice cream and bread, where the foam phase also provides an essential contribution to the sensory properties of the food, foam stability needs

to be maintained for considerably longer. Generally, to avoid separation and destabilization, immobilization of bubbles can be achieved through modification of the continuous phase rheological properties (e.g., increasing viscosity, creating a yield stress or completely solidifying the continuous phase).

Gels

The formation of a gel state usually occurs as a consequence of the formation of bonds between interacting molecules or colloidal particles within a solvent medium. This can lead to the establishment of a percolating structure that is essentially able to immobilize the solvent phase, transitioning a material from a sol state to a gel state. Gels are often considered elastic materials, and while this is certainly true for a number of examples (such as boiled egg white), it is important to note that depending on the type and extent of interactions, significant diversity in gel material/textural properties can be generated. Accordingly, structures can be produced with material properties varying from soft, readily yielding materials (such as yogurt), through to firm, elastic materials (jelly) and hard, fracturable materials (such as aged cheddar cheese). Gels can also be formed from a variety of food ingredients, based on the broad premise that these materials are able to form bonds, either with each other, or with other ingredients. Specific gelling materials are considered as follows.

Protein gels – Proteins with globular molecular structures (such as whey and soy proteins) can form gels as a consequence of protein denaturation (through a variety of mechanisms, such as temperature, pressure, and solvent conditions) usually followed by formation of covalent bonds between neighboring molecules via disulfide exchange. Gel strength can be influenced by protein concentration, ionic and pH conditions and the interactions with other food ingredients.

Proteins can also form noncovalent gels as a consequence of other forms of interactions. Two common examples include casein micelles in milk (which can form particle gels as a consequence of either acidification or rennet treatment) and gelatine, which can form thermoreversible gels as a consequence of hydrogen bond formation.

Polysaccharide gels – A number of polysaccharides are able to undergo gelation. This can be achieved through varying mechanisms of interaction including hydrogen bonding (such as for agar), electrostatic cross-linking (low methoxylated pectin, sodium alginate in combination with calcium ions), and hydrophobic interactions (hydroxymethyl cellulose). Gel properties can be influenced as a consequence of polysaccharide concentration, the number and availability of junction sites, and by solvent conditions. Combinations of polysaccharides can generate some interesting synergistic behaviors, allowing expansion of the range of material/textural properties achievable, and similarly interactions with other ingredients (e.g., the interaction between k-carrageenan and milk) can also create specific pathways for gelation.

Emulsion-based gels – The interfacial layer of emulsion droplets can allow droplet–droplet or droplet–continuous phase interactions to take place. Where droplets are able to stick together (again, a number of mechanisms may allow interaction to occur), network structures can be formed that percolate the system, causing gelation to take place. Gel material properties are dependent on the type and strength of interaction, the amount of fat present, and the influence of the continuous solvent phase (and any other materials present within it). Cream cheese provides an example of such a structure.

Suspensions

A suspension is a colloidal state in which a solid phase is dispersed in a liquid. This particular state can be found in a number of food products such as cake batters, molten chocolate, salad dressings, and chocolate milks. For suspensions, the dispersed phase is generally of higher density than the surrounding continuous phase and (where the continuous phase remains liquid) will tend to sediment over time. To ensure that a solid dispersed phase remains uniformly distributed within a food often requires some form of kinetic trapping, usually as a consequence of altering continuous phase rheology such that the solid particles become locked into the structure.

In this way, sugar particles dispersed in the molten fat phase of chocolate are fixed in place on solidification of the chocolate on cooling. For a product such as chocolate milk, solidification of the continuous phase to immobilize cocoa particles would not be appropriate for the product. However, the implementation of a very weak yield stress (through the addition of the polysaccharide carrageenan) can be sufficient to inhibit sedimentation (while still allowing the liquid to properly flow during pouring and consumption). A similar approach is taken with some salad dressings, where xanthan (another polysaccharide) can be used to impart a weak yield stress to arrest the sedimentation of added herbs.

Food Rheology

Food rheology is the study of the consistency and flow of food under specified applied forces, to understand the underlying physicochemical principles of ‘structuring’ by food materials and their interaction.

The degree of fluidity, consistency and other mechanical properties are important in determining how food ingredients can be mixed and processed during food manufacturing, what kind of product texture can be achieved, and how stable it will remain. The application of rheology to study food is primarily for the following three purposes (Fischer and Windhab, 2011). First, food engineers, who try to develop rheology–process relationships of the food, use rheological data for process or product optimization. Typical flow processes in food processing include mixing/stirring, dispersing, extrusion, coating, injection molding, and spraying.

Rheological measurements are also used in analytical to semiempirical modeling as well as in numerical flow process simulations. Second, material scientists or physicists focus on rheology–structure relationships of soft materials. Model food systems are often used to establish the material properties and develop science for new formulations and food properties. Last, food technologists aim at comparative characterization of food products to establish rheology–property relationships. Typical ‘properties’ of interest correlated with rheology are (1) sensory/perception characteristics (e.g., texture), (2) stability, (3) convenience aspects (e.g., portioning, scooping, dosing, filling), and (4) nutritive characteristics (e.g., release kinetics, satiety).

Rheological characterization involves quantification of the functional relationships between deformation, stresses, and the resulting rheological properties such as viscosity, elasticity or viscoelasticity, flow behavior, and recovery (Fischer et al., 2009). In principle, the experiments are usually carried out by either imposing a small force (*stress*) and measuring the deformation of the sample (*strain*), or a fixed amount of movement (*strain*) and measuring the *stress* developed in the sample. The small deformation measurements enable materials to be probed over supermolecular distances (e.g., nanometer to micrometer range) to establish the relationships between levels of structures and structural organization. On the other hand, measurements made under conditions of large deformation can provide information complementary to that obtained in the small deformation range, particularly time-dependent and nonlinear viscoelastic behavior at large strains and stresses. Measurements obtained at a large deformation regime are more relevant to the use of food materials in practice and the eating experience such as cutting, spreading, or chewing.

Food Texture

Food texture plays an important role in whether the consumers like the food product or not (Foegeding et al., 2011). Texture is one of the attributes used by consumers to assess the food quality. Food texture is part of our sense when we feel the food in our mouth. It is can be described in the terms such as ‘hard,’ ‘soft,’ ‘liquid,’ ‘solid,’ ‘rough,’ ‘smooth,’ ‘creamy,’ ‘crumbly,’ ‘crispy,’ ‘lumpy,’ ‘gritty,’ etc. Such textural terms are directly related to the density, viscosity, surface tension, and other physical properties of a particular food product (Pascua et al., 2013).

Ultimately, the textural characteristics of a food are measured by sensory assessment tests. Since texture is inextricably linked to food structures at micro- and macrolevels and are strongly influenced by the interactions of food biopolymers such as proteins, polysaccharides, and lipids, instrumental methods designed to measure rheological and/or mechanical properties may be used to establish parameters that relate to relevant sensory textural characteristics (Chen and Opara, 2013).

In addition to the direct contribution to consumer acceptance, texture also has a vital secondary role on modulating flavor release and perception. If flavor components in a food are to be sensed, they must be released from the food matrix to reach the appropriate taste receptors. This release of flavor is closely associated with the way in which the food structure breaks down in the mouth relating to both the initial texture of the food and the change in texture throughout mastication (Stieger and Van de Velde, 2013).

Food Breakdown

The structure of any given food product is essentially a template of the properties of that food system, representing its material behavior, physical stability, appearance, sensory performance, and other attributes related to the overall quality of that particular food. For manufactured foods, this is primarily considered for product structures essentially in the final processed and packaged form; in other words, a snapshot of the structure as it sits on the supermarket shelf. However, if one wishes to develop correlations between food structure and aspects such as technical functionality (e.g., the spreadability of butter), sensory, and even behavior during digestion, then it becomes essential to determine the structural changes taking place for any given food in response to the environmental conditions that are applied to that food during use, consumption, or digestion.

Thus, the sensory properties of any food can be established as a consequence of the in-mouth failure of that material in response to the conditions of shear generated by the teeth and movement within the oral cavity, by the interaction and dilution with saliva (including the presence of salivary enzymes), by the temperature in mouth, and of course the residence time needed to render the food into an altered structural state appropriate for swallowing (Chen, 2015). The investigation of structural dynamics in the mouth is termed oral processing and is a rapidly growing branch of the Food Structure discipline. This is facilitated by the development of increasingly sophisticated models able to replicate the oral cavity, particularly regarding the shear forces present during mastication, and which allow the changes in food structure during mechanical simulation of the eating process. Additionally, *in situ* analysis of structure changes during eating can be achieved through techniques such as articulo-graphy, which allows the breakdown of foods during eating to be directly related to the temporal development of sensory attributes. This approach can be applied to virtually any food type.

Similarly, the study of food material breakdown during gastrointestinal transit is increasingly being used to demonstrate the role of food structure in the digestion and uptake of micro- and macronutrients (Norton et al., 2014). This has been a rapidly developing field of research, aided by the development of sophisticated static and dynamic *in vitro* models that can increasingly be considered representative of digestive physiology (particularly with regard to pH, biochemical, and mechanical conditions) within the stomach and small intestine. Such models are able to determine how the digestion key nutrients, such as proteins and lipids, are influenced by the response of food materials to the digestive environment. An understanding of this relationship can allow food structures to

be developed with the ability to impart specific digestive properties, providing appropriate validation using human studies has been carried out (Bornhorst and Singh, 2014).

Conclusion

The quality of a food product, in terms of sensory, shelf life, appearance, or nutritional value, is governed by the structure of that food. An understanding of how ingredients are assembled through processing, and how structural elements are across diverse length scales, impacts the material and technical properties of foods and can provide a considerably more iterative means of manipulating product properties. Thus, the intrinsic functionality of raw materials, ingredients, and additives can be tailored to specifically deliver a requisite structural contribution to any given food product.

More recently this approach has been extended to relate the structural disassembly of foods. Knowledge of how foods break down in the mouth through to the human digestive system provides new insights as to how the dynamics of product microstructure influence both food perception, digestion and nutrient absorption. Such knowledge will ultimately allow for the design and development of products able to meet specific requirements in terms of their properties, thus enhancing product value, whether it is in terms of cost, quality, or nutritional benefit.

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